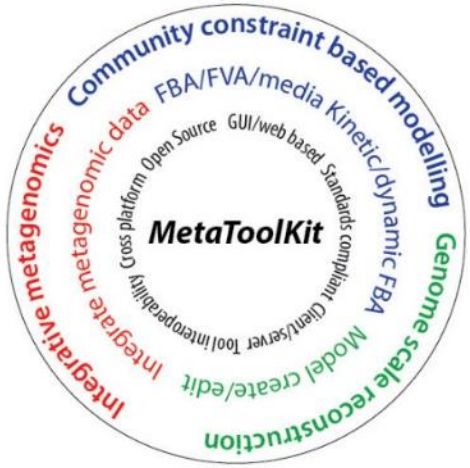
MetaDraft

Building models at genome scale



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Systems Bioinformatics

VU University Amsterdam

BE-Basic Foundation

F08.005.01

CHR HANSEN Improving food & health

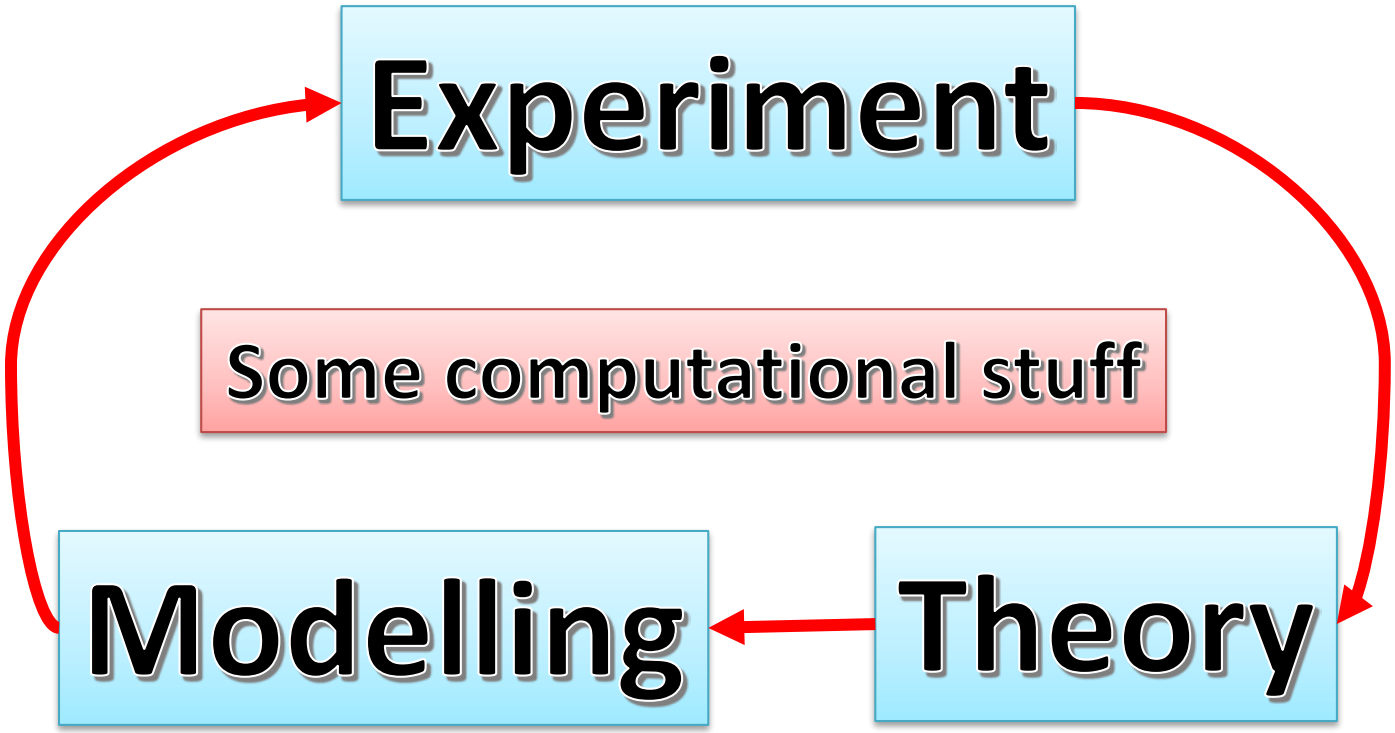
Corbion

DSM BRIGHT SCIENCE. BRIGHTER LIVING.

NIOO

vrije Universiteit amsterdam

Systems biology and modelling



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Constraint-based, genome-scale modelling

Research article

Highly accessed

Genome-scale modeling using flux ratio constraints to enable metabolic engineering of clostridial metabolism in silico

Michael J McAnulty, Jiun Y Yen, Benjamin G Freedman and Ryan S Senger

Research article

Highly accessed

Open Acces

Metabolic network analysis predicts efficacy of FDA-approved drugs targeting the causative agent of a neglected tropical disease

Arvind K Chavali[1], Anna S Blazier[1], Jose L Tlaxca[1], Paul A Jensen[1], Richard D Pearson[2,3] and Jason A Papin[1,4]*

Research article

Highly accessed

Open Ac

Genome-scale metabolic reconstructions of *Pichia stipitis* and *Pichia pastoris* and *in silico* evaluation of their potentials

Luis Caspeta†, Saeed Shoaie†, Rasmus Agren, Intawat Nookaew and Jens Nielsen*

AMERICAN SOCIETY FOR MICROBIOLOGY | Applied and Environmental Microbiology

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Institution: VRIJE UNIVERSITEIT

Using a genome-scale metabolic model of *Enterococcus faecalis* V583 to assess amino acid uptake and its impact on central metabolism

Nadine Veith[1#], Margrete Solheim[2], Koen W. A. van Grinsven[3], Brett G. Olivier[4], Jennifer Levering[1], Ruth Großeholz[1], Jeroen Hugenholtz[3], Helge Holo[2], Ingolf Nes[2], Bas Teusink[4] and Ursula Kummer[1]

This Article

Accepted manuscript posted online 19 December 2014, doi: 10.1128/AEM.03279-14

AEM.03279-14

» Abstract

PDF

Supplemental material

Article Usage Stats

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PLOS ONE

Community Flux Balance Analysis for Microbial Consortia at Balanced Growth

Ruchir A. Khandelwal[1,2], Brett G. Olivier[2,3], Wilfred F. M. Röling[1,3], Bas Teusink[2,3], Frank J. Bruggeman[2,3]*

OPEN ACCESS Freely available online

PLOS ONE

Multiscale Modeling of Metabolism and Macromolecular Synthesis in *E. coli* and Its Application to the Evolution of Codon Usage

Ines Thiele[1,2,3]*, Ronan M. T. Fleming[1,4], Richard Que[3], Aarash Bordbar[3], Dinh Diep[3], Bernhard O. Palsson[3]

Molecular BioSystems

RSC Publishing

REVIEW

View Article Online

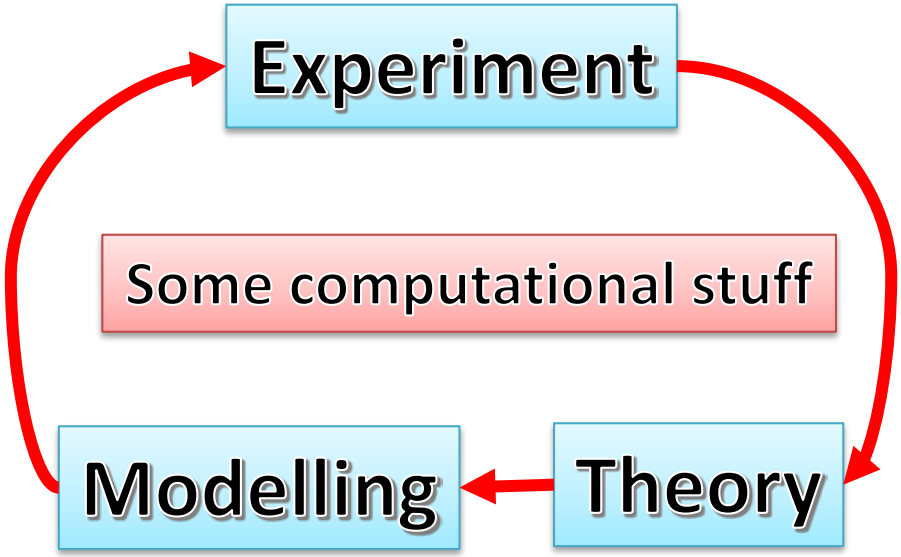
View Journal | View Issue

Cite this: *Mol. BioSyst.*, 2013, 9, 167

Analysis of omics data with genome-scale models of metabolism

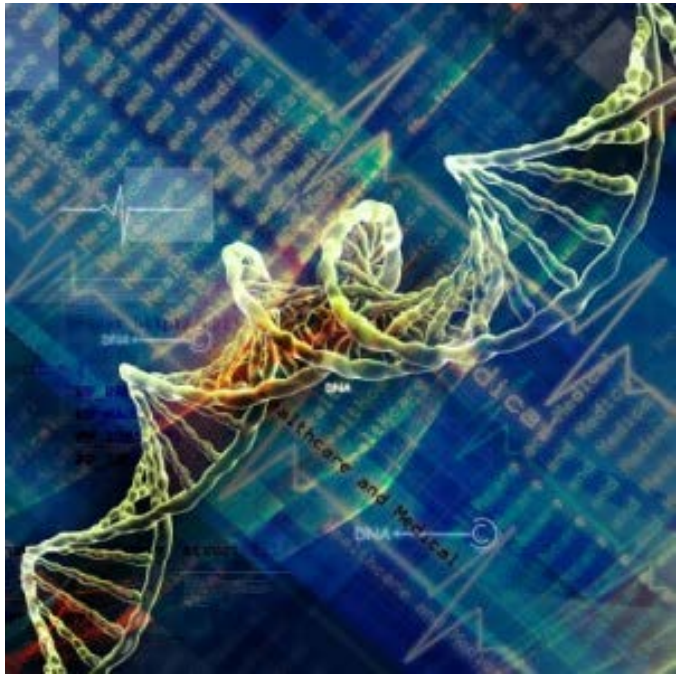
Daniel R. Hyduke,*[a] Nathan E. Lewis[ab] and Bernhard Ø. Palsson[a]

Why isn’t everyone doing this?



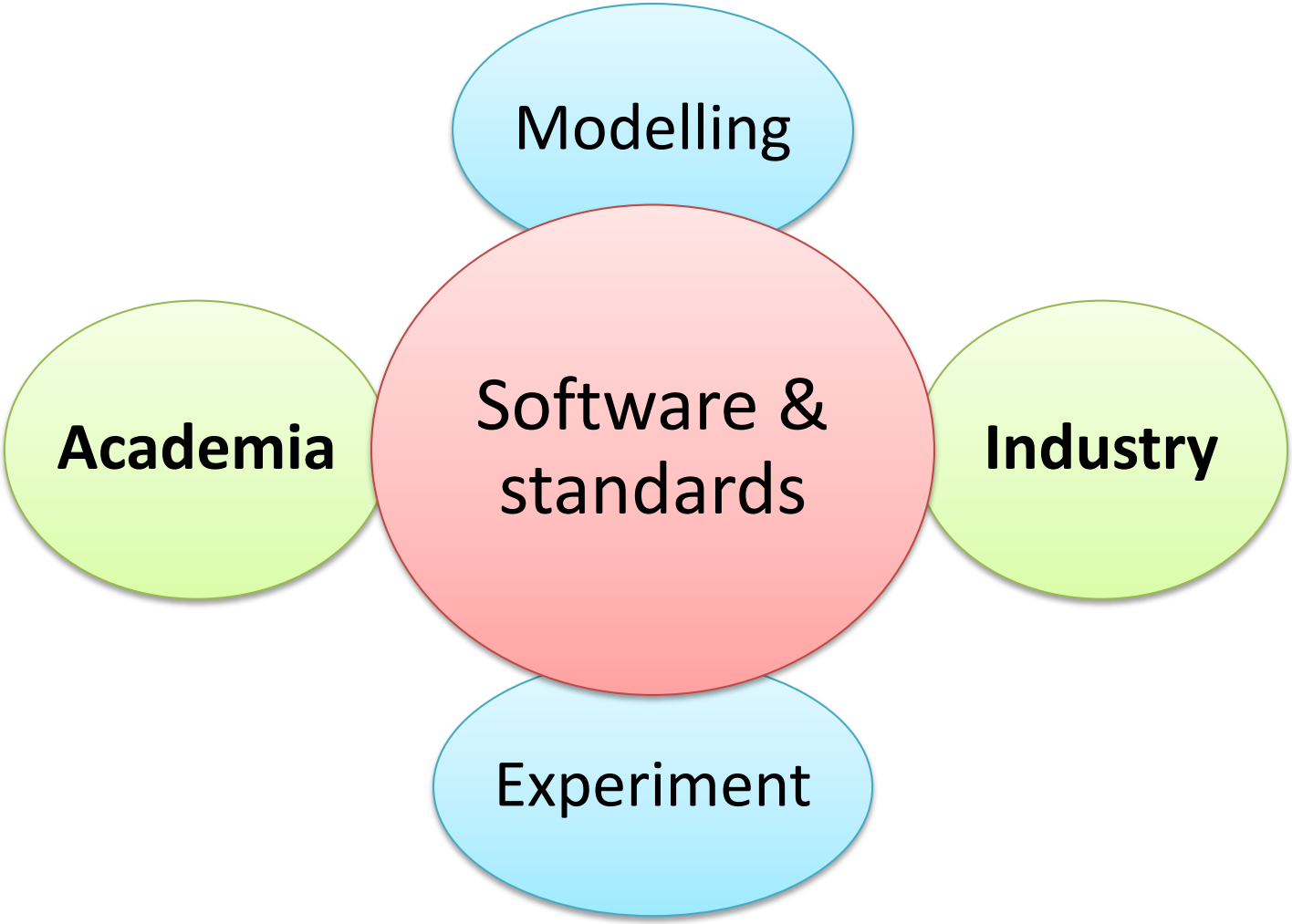
Some computational stuff





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One solution: build better academic software



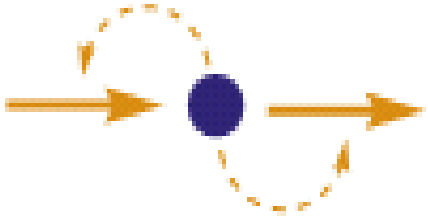
A small case study

FiJo



Reconstruction of genome scale metabolic models

CBMPy



Interactive constraint-based modelling framework

cbmpy.sf.net

FAME

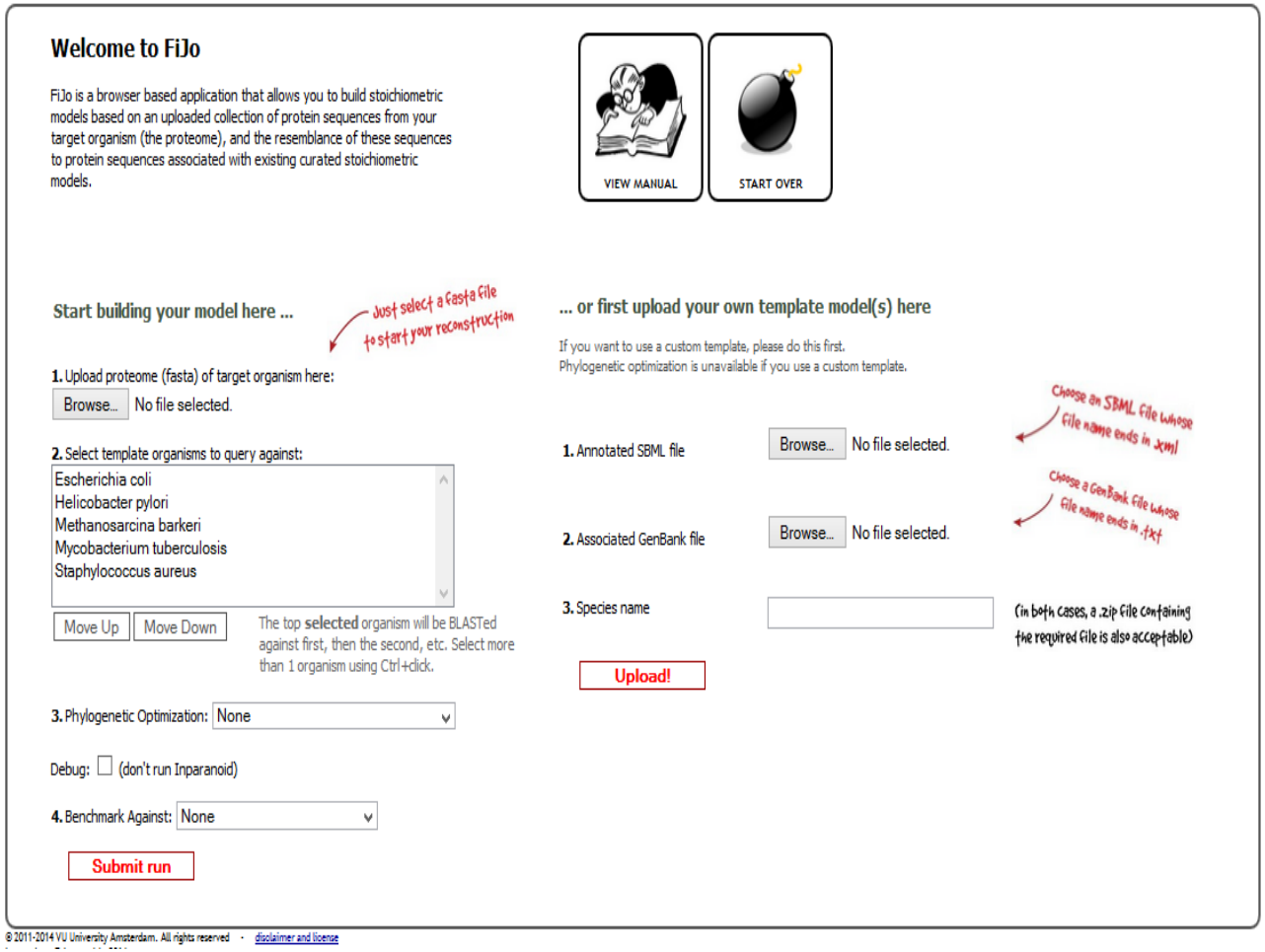


Online FBA, model editor and visualisation tool

f-a-m-e.org

- Research software developed at the VU
- All projects are Open Source
- Covers the entire GSR model creation/analysis cycle

Model reconstruction: FiJo



Developed by Joost Boele and Filipe Branco dos Santos

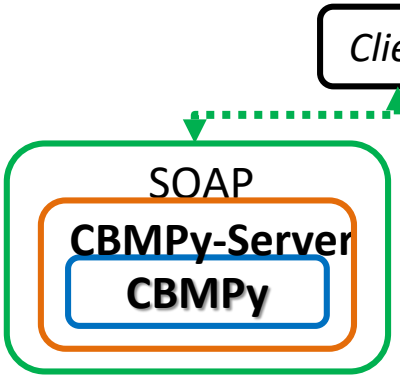
- Input is FASTA
- Generates a set of candidate reactions, metabolites and compartments
- Outputs SBML

- Web-based
- No session tracking
- No annotation
- Unmaintained since 2014

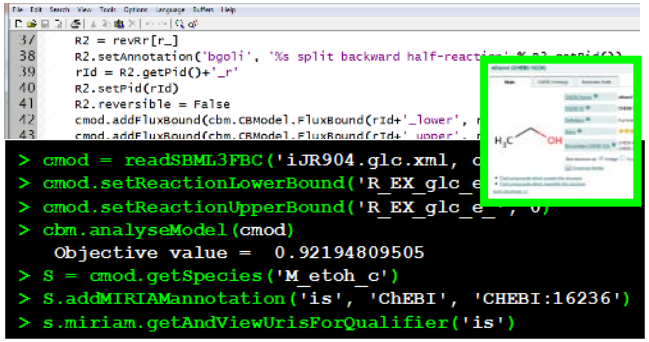
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CBMPy

http://cbmpy.sourceforge.net



Web services

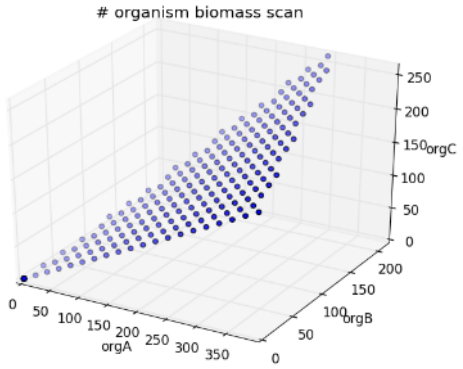


Advanced scripting language

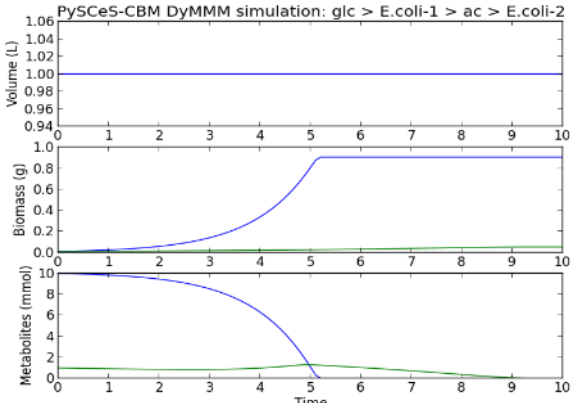
CBMPy
PySCeS Constraint Based Modelling

SBML3 + FBC, RDF and KeyValueData annotation

- Model definition in all SBML versions, Advanced analysis
- Extendible design
- Outputs SBML, Excel, COMBINE
- Supports MIRIAM & custom annotation



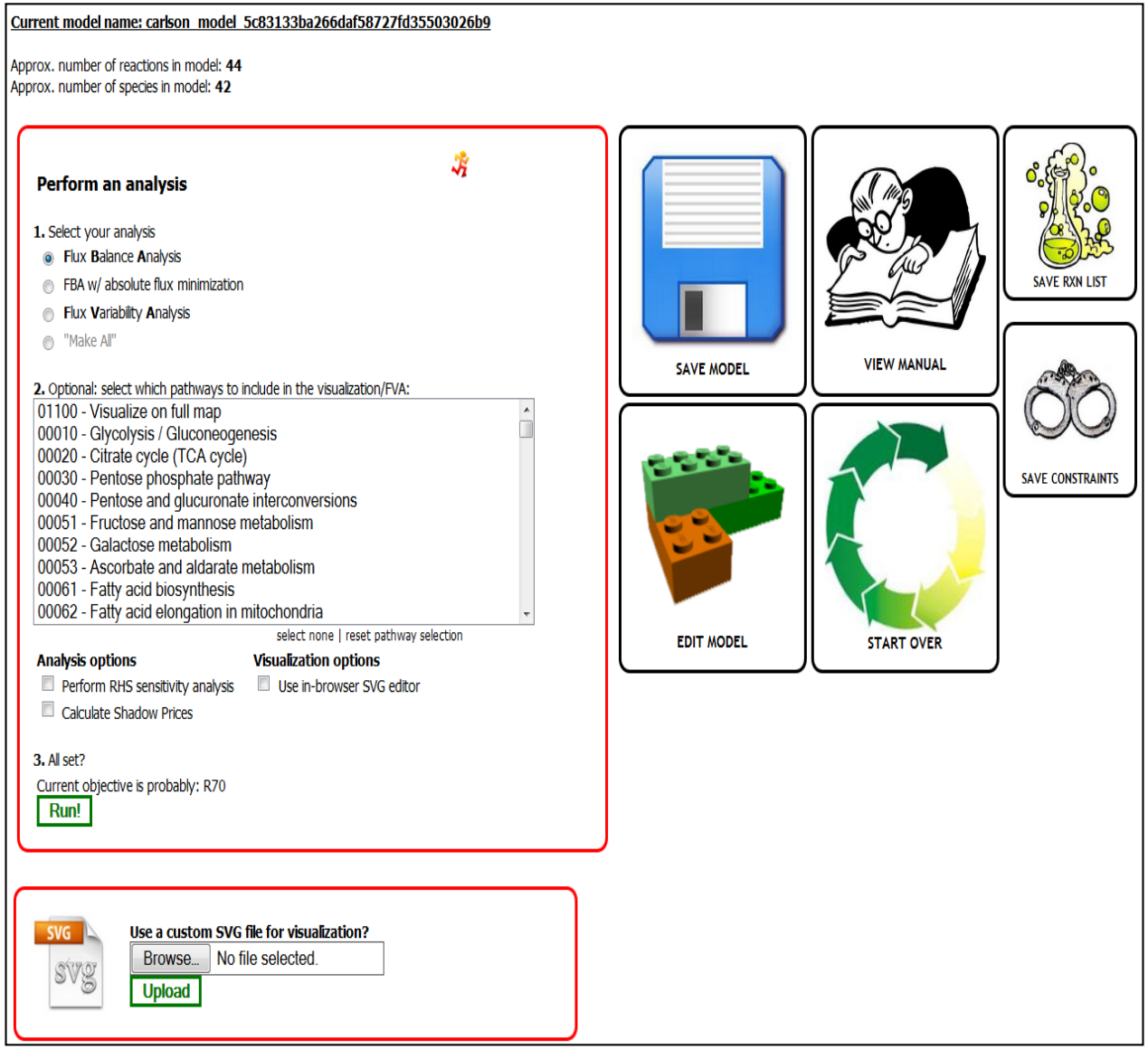
Static cFBA analysis



Dynamic multispecies dFBA

- Standalone
- Console based
- Not “user” friendly
- No session tracking
- Actively developed

FAME (f-a-m-e.org)



- Input is SBML
- Basic analysis
- Result visualization
- Uses CBMPy
- Outputs SBML

- Web based
- Difficult to extend
- No annotation
- Maintained but no new features since 2014

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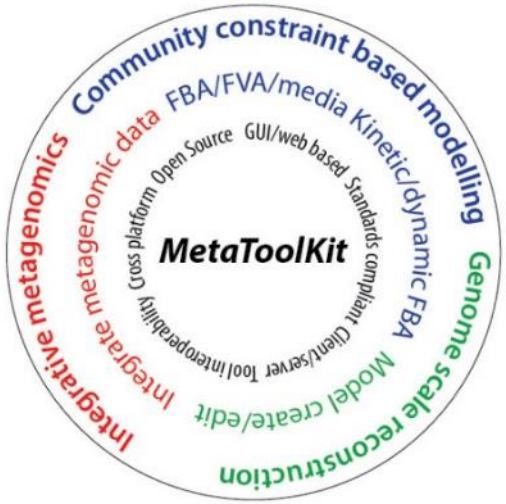
Case study: not too bad

Capability	Support		
	FiJo	CBMPy	FAME
• Accessible	✓	✗	✓
• User friendly	✗	✗	✓
• Web-based	✓	✗	✓
• Flexible/extendible	✗	✓	✗
• Advanced functionality	✗	✓	✗
• Easy (local) installation	✗	✓	✗
• Industry ICT ready	✗	✗	✗
• Standards support	✓	✓	✓
• Open source,	✓	✓	✓
• Actively maintained	✗	✓	✓

MetaToolkit

Building and analysing models at genome scale

MetaDraft, MetaMod, MetaSim



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MetaToolKit

- **For the creation and analysis of genome-scale metabolic models**
  - Integrates the **best parts of FiJo/CBMPy/FAME**
  - **Built with industrial** BE-BASIC partners
- **Integrated platform suitable for use in an industrial environment**
  - No **dependency** on commercial products e.g. MATLAB, Mathematica …
  - Cross platform (**Windows** and optionally Linux/Mac)
  - **Standalone software**, no off network or web based tools
  - **Advanced user interfaces** (graphical) and data visualisation capabilities
  - **Model annotation** and **provenance** leveraging open **standards**
  - Integration into **enterprise IT** infrastructures
- **Development driven by partner needs**

MetaToolKit partners

Chr. Hansen

Reconstruction and media generation

CORBION

Reconstruction and analysis

DSM

Dynamic analysis

NIOO-KNAW

Ecosystem reconstruction and analysis

Related projects within BE-Basic

Flagship 8

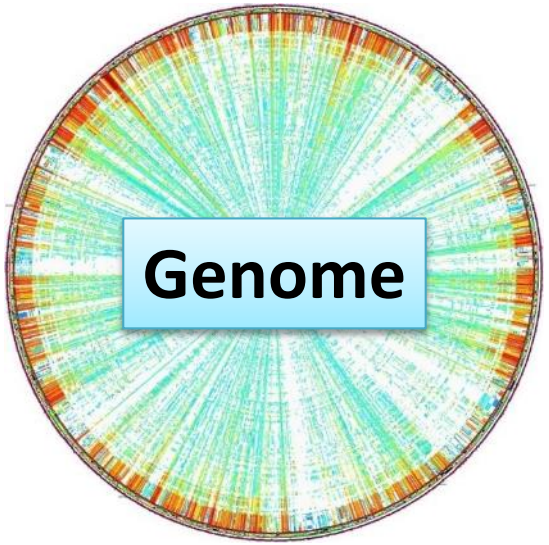
- F08.004.01 MicroControl

Flagship 7

- F07.003.02 Intelligent mining

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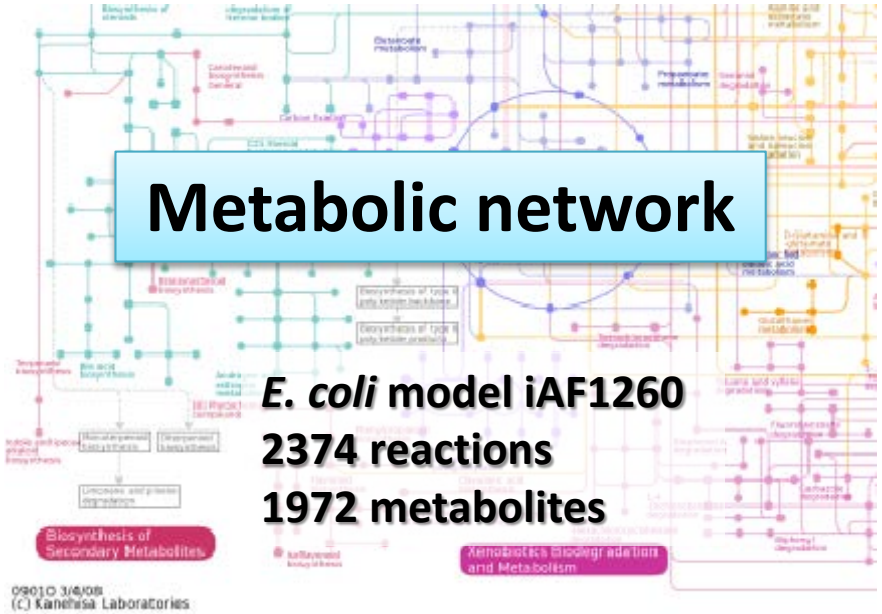
Reconstructing a genome-scale model



ATCG

Genome annotation

- evidence for gene
- protein definition
- reaction stoichiometry
- compartmentalization
- chemical/charge balancing



$B + 2\,C \rightarrow D$

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Reconstruction pipelines

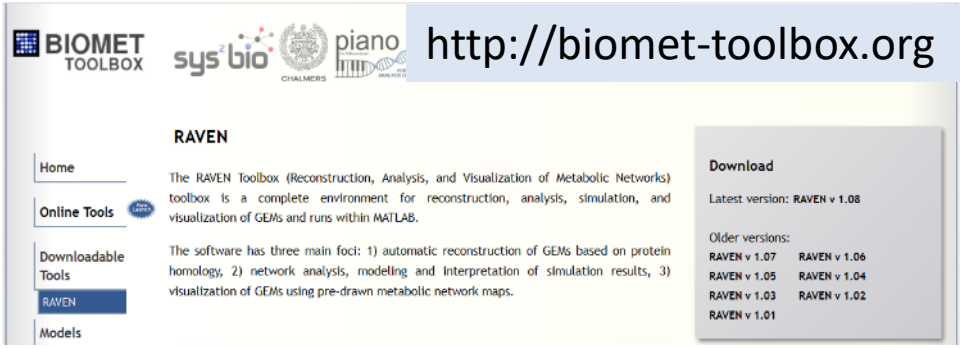
http://modelseed.org/



ModelSEED

- Online only
- RAST annotation engine (bi-directional best hit on internal database)
- Exports COBRA SBML (no annotation)

http://biomet-toolbox.org

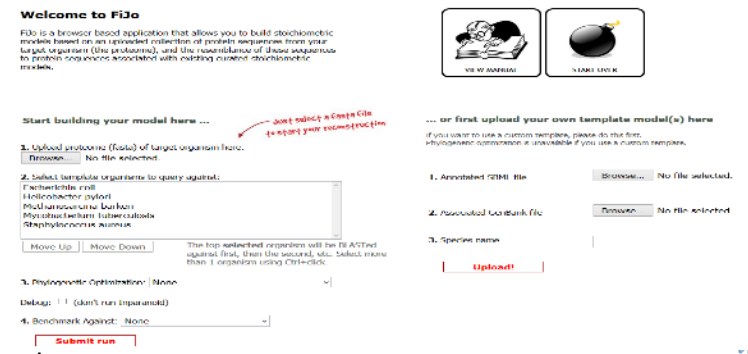


RAVEN

- MATLAB only
- bi-directional BLASTp template models or KeGG (pre 2013)
- Exports COBRA SBML

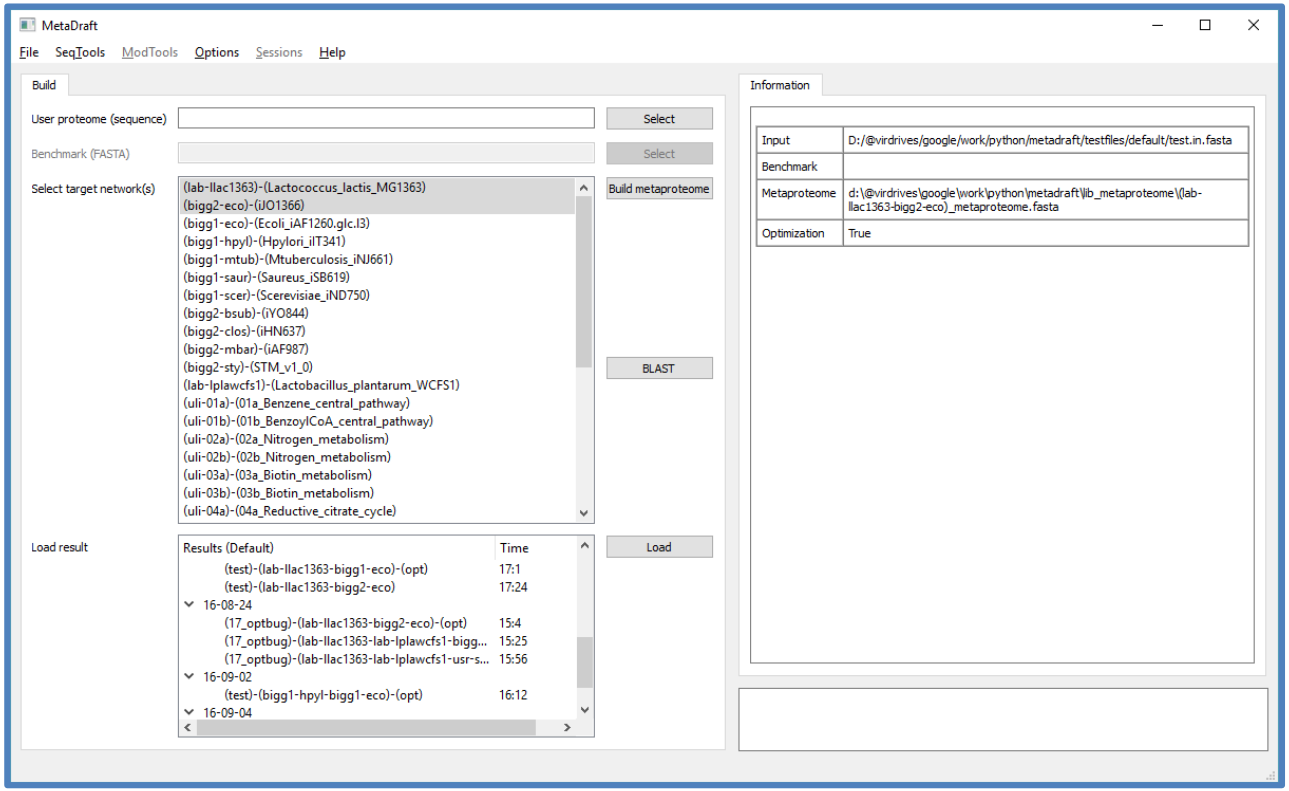
FiJO

- Online only
- Based on autograph
- Orthology based approach
- Exports SBML3 FBC V1



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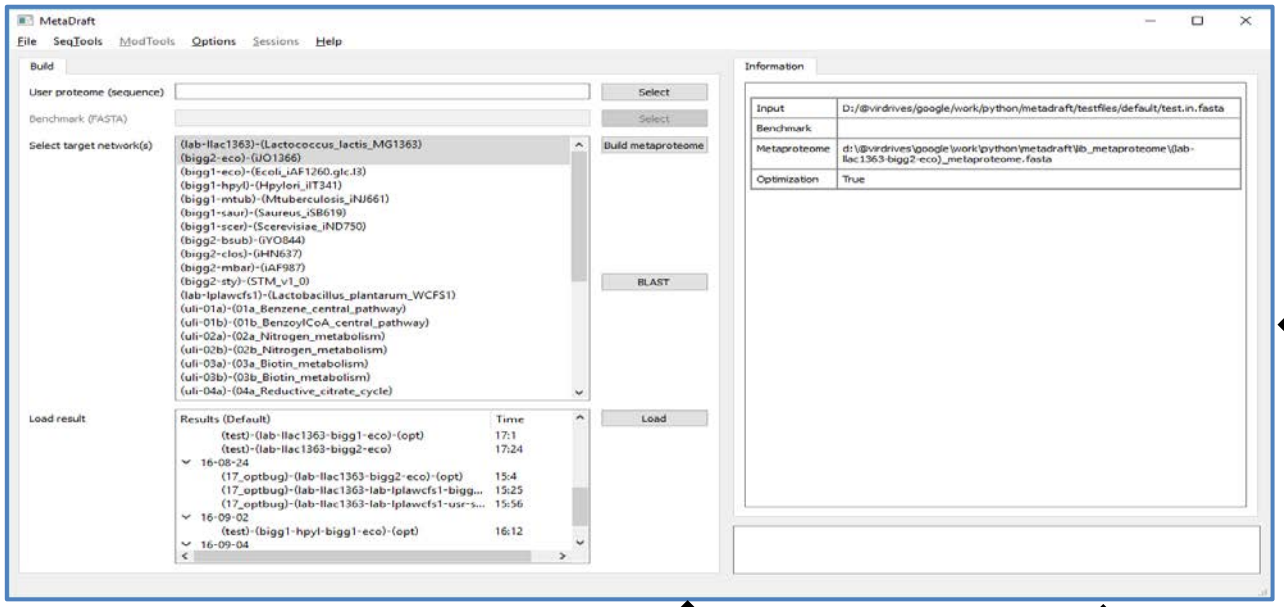
MetaDraft: model reconstruction



- Uses the **Autograph** approach and searches for **orthologous** genes
- Generates a set of candidate **reactions**, **metabolites** and **compartments**
- **Maintains** all gene and model **annotation** in output
- **Multi-session support,** provenance tracking and reporting
- **Standalone, cross-platform**, tested on Windows 10, Ubuntu 16.04 and OS X
- Supports latest **SBML Level 3, MIRIAM** and **COMBINE** standards

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MetaDraft



Core technologies

- Python 2.7
- Qt 4.8
- SQLite
- CBMPy
- libSBML
- biopython
- biotools

SeqPlus integration

Models with GPR associations

Saureus_iSB619.xml

iSB619-NC_002745.gbk

Genome with CDS information

SeqPlus models

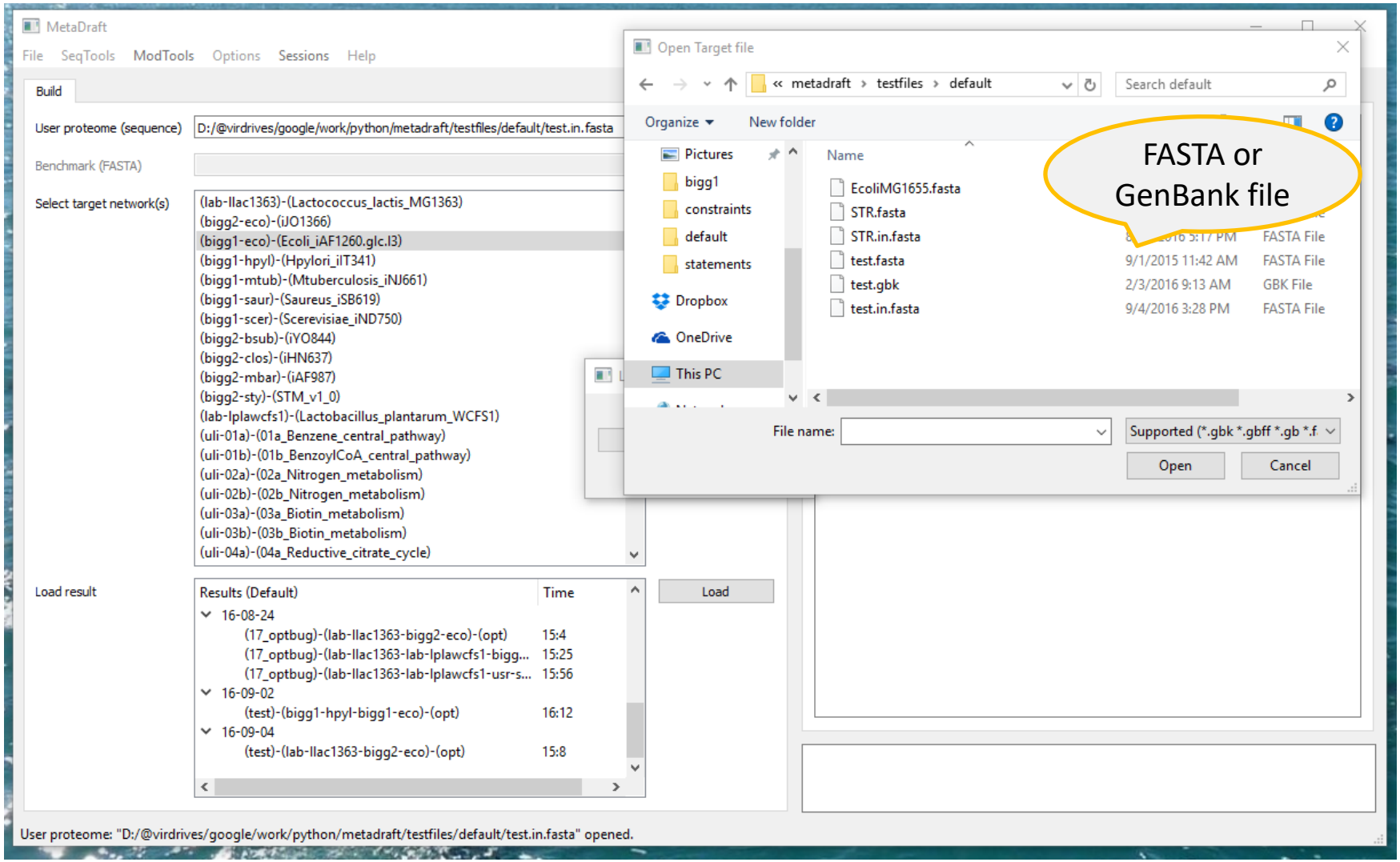
BiGG, BiGG2
EstherDB
Custom models

(bigg1-saur)-(Saureus_isB619)

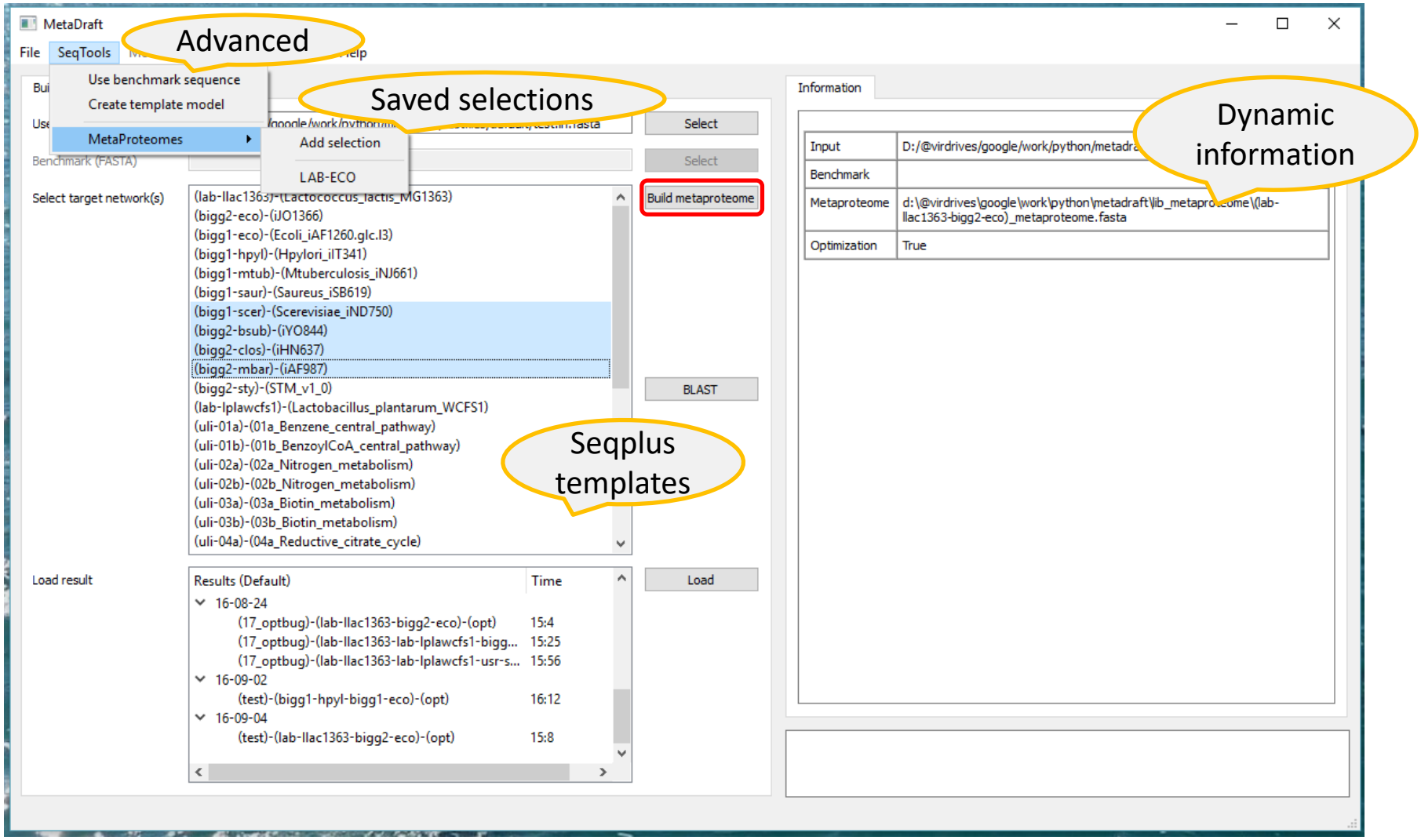
Gene annotation

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Input sequences

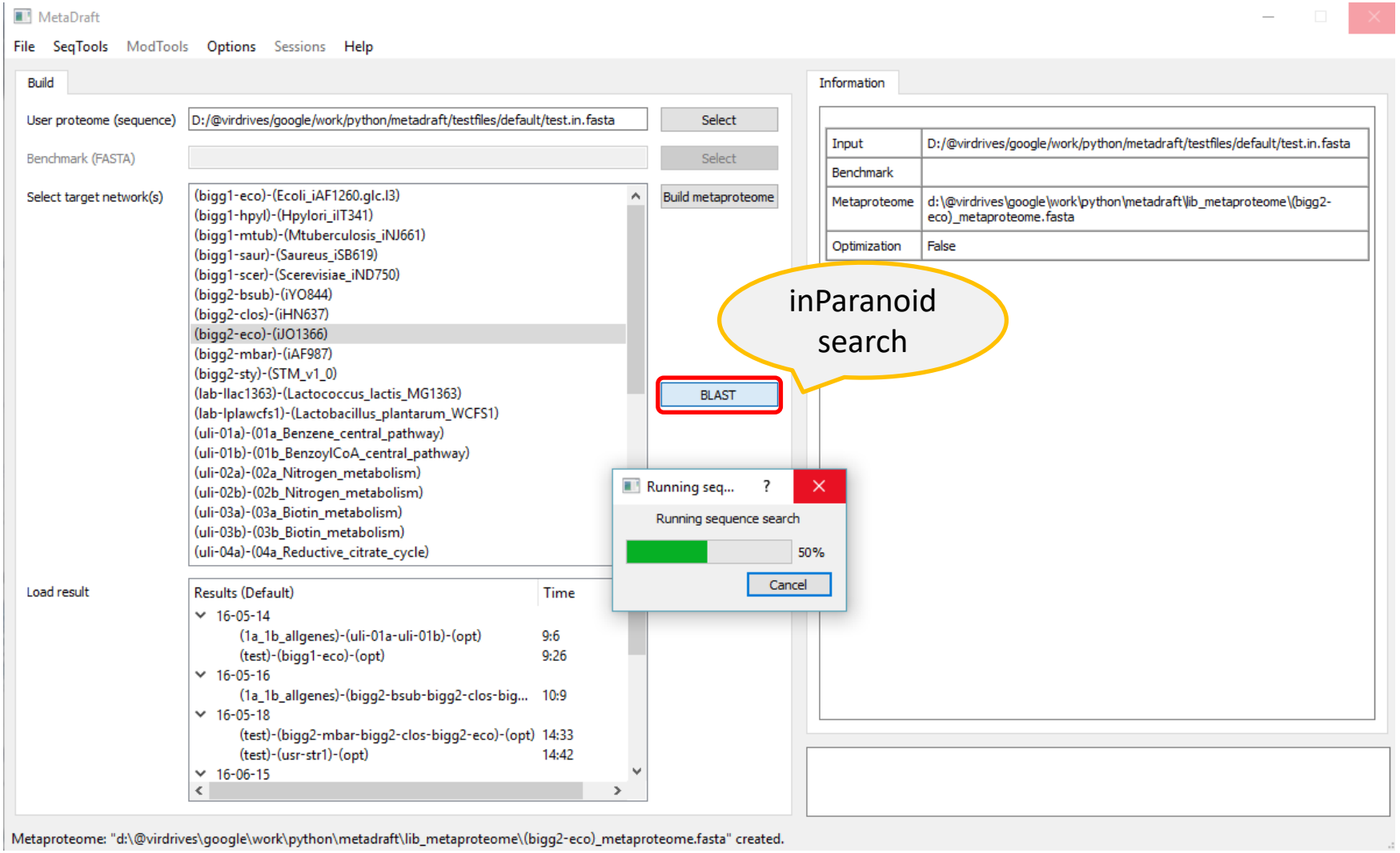


Build Metaproteome



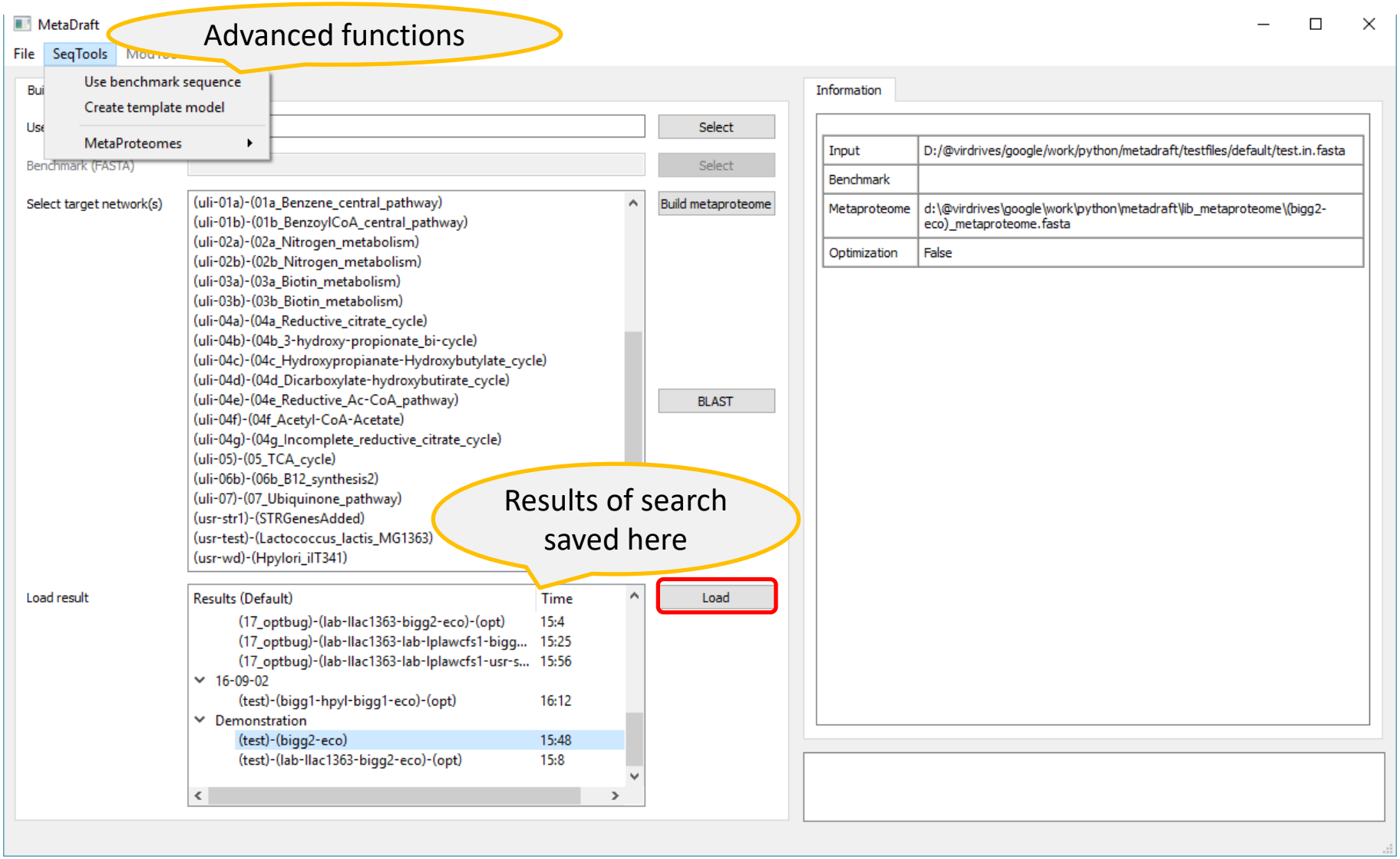
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Search/BLAST



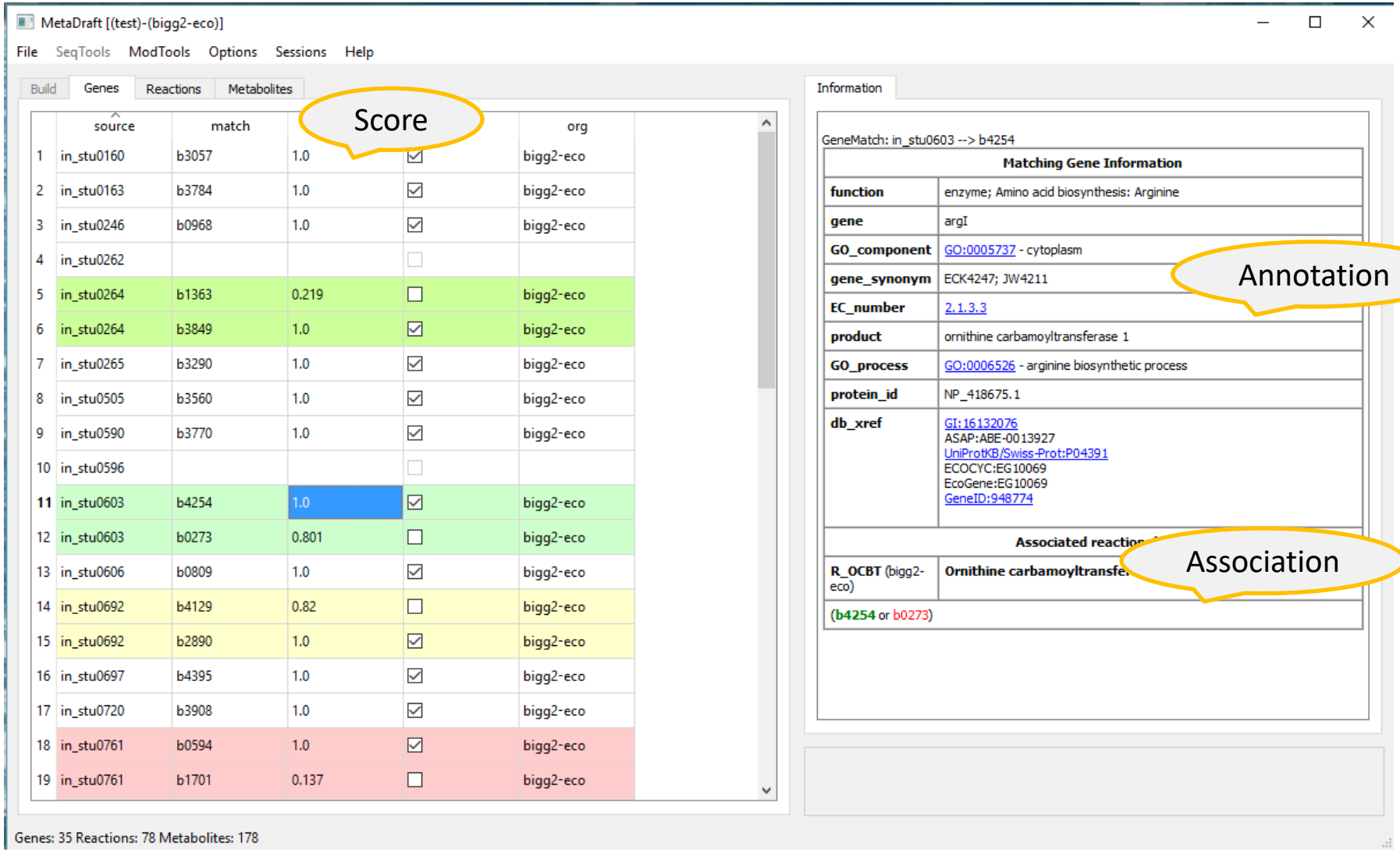
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Load results



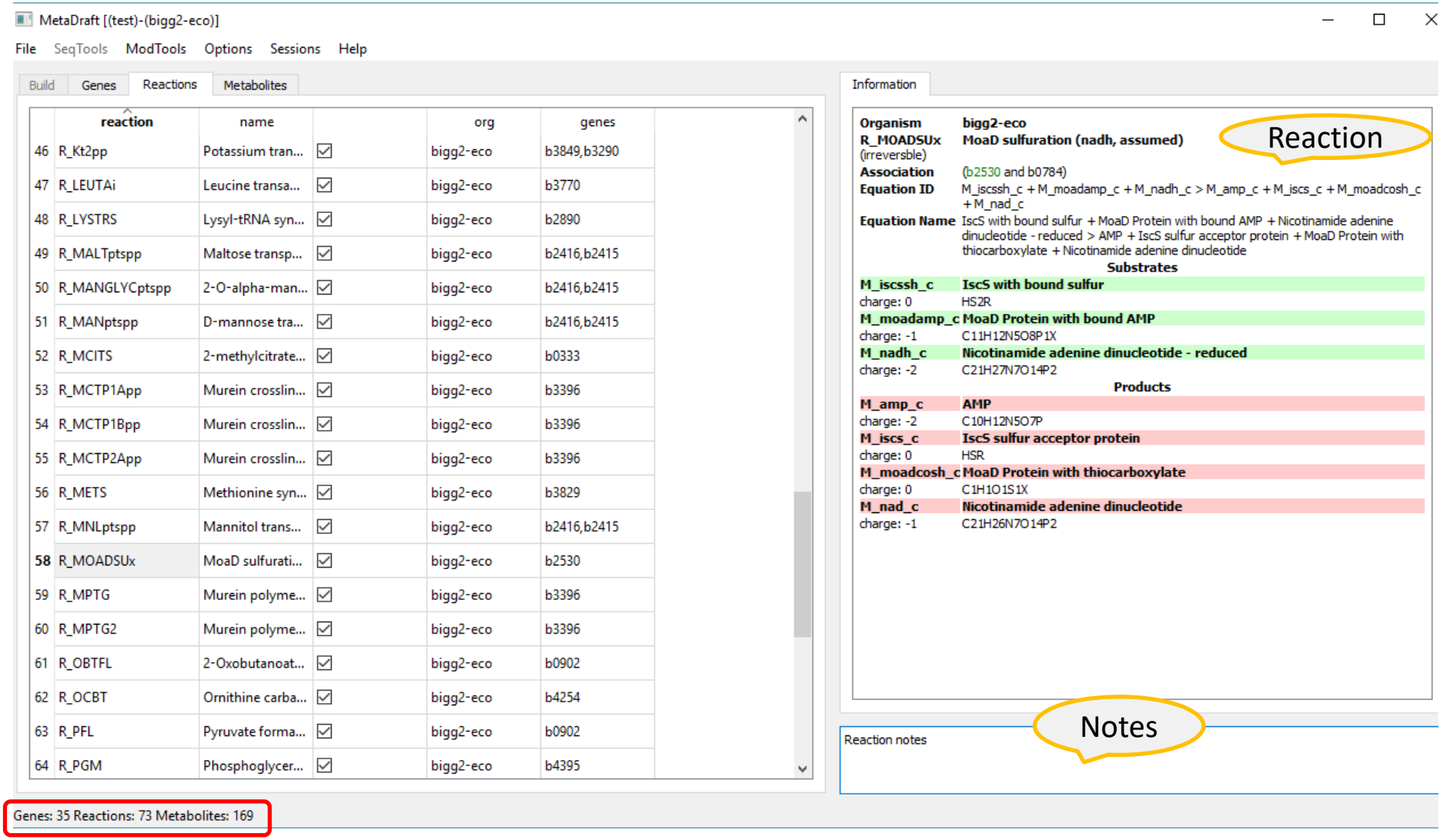
b.g.olivier@vu.nl

Select genes



	source	match			org
1	in_stu0160	b3057	1.0	☑	bigg2-eco
2	in_stu0163	b3784	1.0	☑	bigg2-eco
3	in_stu0246	b0968	1.0	☑	bigg2-eco
4	in_stu0262			☐	
5	in_stu0264	b1363	0.219	☐	bigg2-eco
6	in_stu0264	b3849	1.0	☑	bigg2-eco
7	in_stu0265	b3290	1.0	☑	bigg2-eco
8	in_stu0505	b3560	1.0	☑	bigg2-eco
9	in_stu0590	b3770	1.0	☑	bigg2-eco
10	in_stu0596			☐	
11	in_stu0603	b4254	1.0	☑	bigg2-eco
12	in_stu0603	b0273	0.801	☐	bigg2-eco
13	in_stu0606	b0809	1.0	☑	bigg2-eco
14	in_stu0692	b4129	0.82	☐	bigg2-eco
15	in_stu0692	b2890	1.0	☑	bigg2-eco
16	in_stu0697	b4395	1.0	☑	bigg2-eco
17	in_stu0720	b3908	1.0	☑	bigg2-eco
18	in_stu0761	b0594	1.0	☑	bigg2-eco
19	in_stu0761	b1701	0.137	☐	bigg2-eco

Select reactions

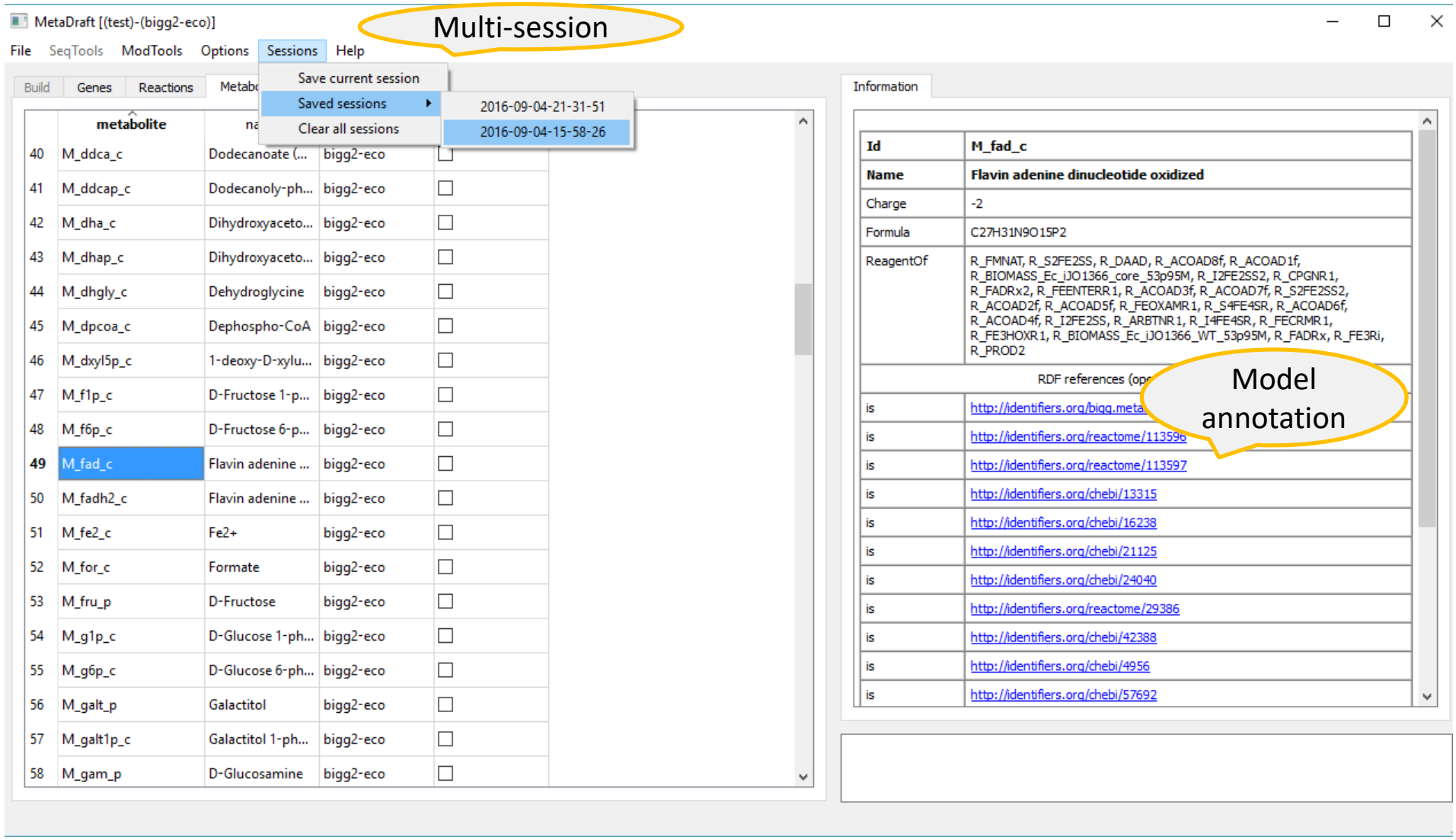


	reaction	name		org	genes
46	R_Kt2pp	Potassium tran...	☑	bigg2-eco	b3849,b3290
47	R_LEUTAi	Leucine transa...	☑	bigg2-eco	b3770
48	R_LYSTRS	Lysyl-tRNA syn...	☑	bigg2-eco	b2890
49	R_MALTptspp	Maltose transp...	☑	bigg2-eco	b2416,b2415
50	R_MANGLYCptspp	2-O-alpha-man...	☑	bigg2-eco	b2416,b2415
51	R_MANptspp	D-mannose tra...	☑	bigg2-eco	b2416,b2415
52	R_MCITS	2-methylcitrate...	☑	bigg2-eco	b0333
53	R_MCTP1App	Murein crosslin...	☑	bigg2-eco	b3396
54	R_MCTP1Bpp	Murein crosslin...	☑	bigg2-eco	b3396
55	R_MCTP2App	Murein crosslin...	☑	bigg2-eco	b3396
56	R_METS	Methionine syn...	☑	bigg2-eco	b3829
57	R_MNLptspp	Mannitol trans...	☑	bigg2-eco	b2416,b2415
58	R_MOADSUx	MoaD sulfurati...	☑	bigg2-eco	b2530
59	R_MPTG	Murein polyme...	☑	bigg2-eco	b3396
60	R_MPTG2	Murein polyme...	☑	bigg2-eco	b3396
61	R_OBTFL	2-Oxobutanoat...	☑	bigg2-eco	b0902
62	R_OCBT	Ornithine carba...	☑	bigg2-eco	b4254
63	R_PFL	Pyruvate forma...	☑	bigg2-eco	b0902
64	R_PGM	Phosphoglycer...	☑	bigg2-eco	b4395

Organism	**bigg2-eco**
R_MOADSUx (irreversble)	**MoaD sulfuration (nadh, assumed)**
Association	(b2530 and b0784)
Equation ID	M_iscssh_c + M_moadamp_c + M_nadh_c > M_amp_c + M_iscs_c + M_moadcosh_c + M_nad_c
Equation Name	IscS with bound sulfur + MoaD Protein with bound AMP + Nicotinamide adenine dinucleotide - reduced > AMP + IscS sulfur acceptor protein + MoaD Protein with thiocarboxylate + Nicotinamide adenine dinucleotide
	Substrates
M_iscssh_c	**IscS with bound sulfur**
charge: 0	HS2R
M_moadamp_c	**MoaD Protein with bound AMP**
charge: -1	C11H12N5O8P1X
M_nadh_c	**Nicotinamide adenine dinucleotide - reduced**
charge: -2	C21H27N7O14P2
	Products
M_amp_c	**AMP**
charge: -2	C10H12N5O7P
M_iscs_c	**IscS sulfur acceptor protein**
charge: 0	HSR
M_moadcosh_c	**MoaD Protein with thiocarboxylate**
charge: 0	C1H1O1S1X
M_nad_c	**Nicotinamide adenine dinucleotide**
charge: -1	C21H26N7O14P2

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Multi-sessions



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MetaDraft in action

CORBION

- Model reconstruction

ChrHansen

- model reconstruction
- generating FASTA meta-proteomes

Ulisses

- Searching for pathways in metagenomics data

Willi

- Model reconstruction

Novel functionality

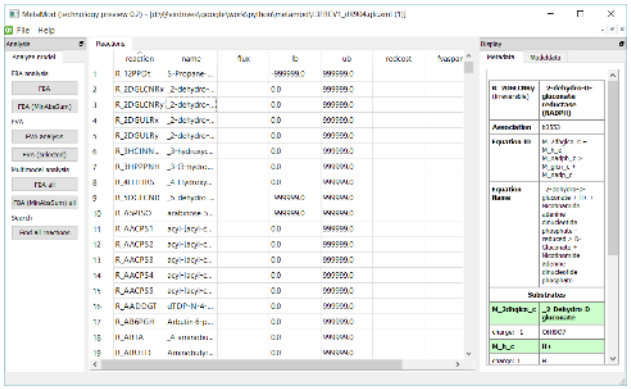
b.g.olivier@vu.nl

MetaToolKit

- **Better quality academic software can mean broader method adoption***
- **Introducing MetaDraft for the creation of genome scale metabolic models (not public, available by request only)**
- **MetaToolkit development continues with MetaMod**

* **Top 10 metrics for life science software good practices**

F1000Research 2016, 5(ELIXIR):2000
doi:10.12688/f1000research.9206.1



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Acknowledgements

Systems Bioinformatics

- Bas Teusink
- Douwe Molenaar

MetaDraft

- Filipe Branco dos Santos
- Willi Gottstein
- Ullises Nunes da Rocha

Software quality metrics

- ELIXIR-EXCELERATE working group
- Software Sustainability Institute

Funding:

- BE-BASIC Foundation
- MetaToolkit partners



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Experiment

Some computational stuff

Modelling

Theory

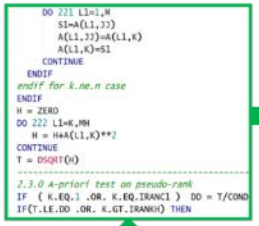
Researchers



Modellers



Developers



MetaToolkit



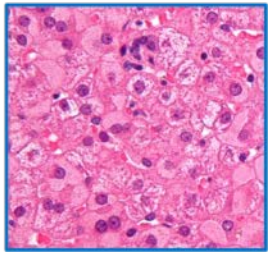












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